

# Coherent Measures of Risk

*Philippe Artzner · Freddy Delbaen · Jean-Marc Eber · David Heath*

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## ABSTRACT

The paper asks what properties a mapping from future net worth to a capital requirement must satisfy to be a defensible measure of risk. Working without a complete-markets assumption, the authors define acceptance sets of future net worths and derive from them four axioms – translation invariance, subadditivity, positive homogeneity, and monotonicity – that any such measure should obey; a measure satisfying all four is called coherent. They show coherence is equivalent to computing the worst-case expected loss over a family of generalized scenarios, which nests the exchange margining system SPAN as a special case. Measured against the axioms, the Securities and Exchange Commission margin rules are shown to be needlessly conservative, and Value at Risk, the industry-standard quantile measure, is shown to violate subadditivity: it can penalize diversification and reward risk transfers that leave every party worse off. The authors propose Tail Conditional Expectation, later known as Expected Shortfall, as a coherent repair that remains close to the VaR methodology regulators already used. The paper does not select a single correct risk measure; it characterizes the admissible class and leaves the choice within it to economic judgment.

## 1. Introduction

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Artzner, Delbaen, Eber, and Heath (hereafter ADEH) address a question that is prior to any specific risk model: before asking how large a loss a portfolio might suffer, one must first ask what it means for a number to be a legitimate measure of that loss in the first place. By 1996, when the working paper began circulating, financial institutions, exchanges, and regulators already possessed several competing devices for this purpose – the Chicago Mercantile Exchange's SPAN margining system, the SEC/NASD margin rules, and the quantile-based Value at Risk (VaR) popularized by J.P. Morgan's RiskMetrics service. None of these had been derived from first principles; each was an engineering solution to a practical problem. ADEH's contribution is to supply the missing first principles: a small set of axioms that any capital-requirement rule should satisfy if it is to support decentralized risk management, and a complete mathematical characterization of the class of rules that satisfy them.

PAPER The central finding is that this class has an exact representation: a risk measure is coherent if and only if it is the supremum of expected loss over a family of probability measures (“generalized scenarios”). This single fact does three things at once. It explains why SPAN – already a scenario-based system – is coherent almost by construction; it exposes the SEC/NASD rules as an unnecessarily conservative special case of the same idea; and it proves that Value at Risk, despite its ubiquity, is not coherent, because it can fail subadditivity: a portfolio's measured risk can exceed the sum of the measured risks of its parts. The paper's applied payoff is Tail Conditional Expectation (TCE), a coherent measure that averages losses in the tail beyond the VaR quantile rather than reporting the quantile itself. Two decades later, its lineal descendant, Expected Shortfall, replaced VaR as the Basel Committee's standard market-risk metric under the Fundamental Review of the Trading Book (FRTB), making this review's subject one of the few pieces of finance theory to have been legislated into the capital requirements of every major bank in the world.

## 2. Research Problem and Hypothesis

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### 2.1 Research Question

The paper does not ask an empirical question. It asks: what set of properties must a mapping  $\rho: G \rightarrow \mathbb{R}$ , from random future net worths to a real-valued capital requirement, satisfy in order to (i) be logically consistent with a well-defined set of “acceptable” future positions, and (ii) support decentralized risk management within a firm or across a regulatory system without perverse incentives? A precise restatement, in the paper's own terms: given an acceptance set of tolerable future net worths, is there a canonical measure of the capital needed to convert an unacceptable position into an acceptable one, and which of the many risk statistics used in practice qualify as instances of that canonical measure?

## 2.2 Motivation

The question matters because capital requirements are not academic abstractions; they are binding constraints with real economic cost, and the rule used to compute them shapes behavior. If a measure fails subadditivity, a regulator has (as the paper's own examples show) created an incentive to concentrate rather than diversify risk, or has allowed two economically identical portfolios – one held whole, one split across legal affiliates – to face different capital charges. If a measure fails monotonicity, a position that dominates another state-by-state could be assessed as riskier. These are not corner cases: the paper demonstrates that VaR, the method endorsed by regulators worldwide by the mid-1990s, fails exactly the property (subadditivity) that decentralized capital allocation depends on. A reviewer should read this paper because it is the origin point of the axiomatic risk-measurement literature that eventually produced Expected Shortfall, convex and spectral risk measures, and the post-2008 regulatory shift away from VaR – and because its critique of VaR is not a modeling refinement but a structural indictment.

## 2.3 Existing Literature

Three practical methodologies pre-date the paper and are its explicit object of analysis. SPAN (Standard Portfolio Analysis of Risk), developed by the Chicago Mercantile Exchange, computes margin as the worst loss across sixteen prescribed scenarios of price and volatility movement. The SEC/NASD rules, adopted by U.S. broker-dealers and mirrored at the Pacific Exchange and the CBOE, compute margin by decomposing an options portfolio into standardized "spread" building blocks and summing prescribed charges. Value at Risk, formalized for practitioners by J.P. Morgan's RiskMetrics Technical Document (1995) and analyzed academically by Duffie and Pan (1997) and the Risk Magazine (1996) special supplement, reports the negative of a chosen quantile of the profit-and-loss distribution. On the theoretical side, the paper builds on Huber's (1981) robust statistics – specifically a representation theorem for sublinear functionals that ADEH's own referee pointed out was mathematically equivalent to their own representation result – and on the convex-analysis fact, already standard in mathematical finance since Merton (1973), that European call prices are convex in strike. An earlier, non-technical version of part of the argument appeared as Artzner, Delbaen, Eber, and Heath (1997), "Thinking Coherently," in Risk Magazine.

## 2.4 Research Gap

What was missing before this paper was not a risk statistic – the literature already had several – but a criterion for judging between them. Existing methods had been justified by regulatory convenience (SEC/NASD), by exchange operational practice (SPAN), or by statistical tractability under normal returns (VaR), never by a stated set of properties that a "good" risk measure should have. Consequently there was no principled way to compare SPAN with the SEC rules, no proof that either was internally consistent, and no formal demonstration that VaR's well-known peculiarities under skewed or fat-tailed distributions reflected an axiomatic failure rather than a modeling nuisance.

## 2.5 Contribution

PAPER ADEH's stated contribution is fourfold: (1) a definition of risk as a random future net worth together with axioms on the set of acceptable such net worths; (2) a definition of the risk measure induced by an acceptance set, and four axioms – translation invariance, subadditivity, positive homogeneity, monotonicity – that together define coherence; (3) a representation theorem proving that coherent risk measures are exactly the scenario-based, worst-case-expectation measures; and (4) an application of this apparatus to show SPAN is (essentially) coherent, the SEC/NASD rules are coherent but needlessly conservative, and VaR is not coherent, followed by the proposal of Tail Conditional Expectation as a coherent alternative. CRITIQUE The claimed contribution is fully demonstrated for the representation theorem and the VaR failure (both proved rigorously with explicit counterexamples), but the paper is more tentative about TCE: it is shown to be coherent and to dominate VaR, but the authors themselves note that TCE is only one member of a large admissible class, chosen for its proximity to existing regulatory practice rather than derived as uniquely optimal. This qualification matters for how much the reader should treat TCE as "the" answer versus one reasonable answer.

## 3. Theoretical Framework and Mechanism

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### 3.1 Primitives: States, Net Worth, and Acceptance

The setting is a single period between dates 0 and  $T$ , over a finite state space  $\Omega$  (so  $G$ , the set of all real-valued functions on  $\Omega$ , is identified with  $\mathbb{R}^n$ ,  $n = \text{card}(\Omega)$ ). A position's risk is represented not by a change in value but by its future net worth  $X \in G$  – a deliberate choice the authors call the "bygones are bygones" principle, since initial cost may not even be defined for over-the-counter positions. A reference instrument carries one unit of date-0 currency into  $r$  units of date- $T$  currency; a default-free zero-coupon bond is the natural example.

The primitive object is the acceptance set  $A \subseteq G$ : the future net worths a regulator, clearing firm, or internal risk manager judges tolerable without extra capital. Four axioms are imposed on  $A$ :

- 2.1 A contains the non-negative cone  $L^+$  – a position that is never negative needs no extra capital.
- 2.2 A does not intersect  $L^{--}$ , the set of net worths negative in every state – a position that is certain to lose money is never acceptable. The strengthened version, Axiom 2.2', requires  $A \cap L^- = \{0\}$ .
- 2.3 A is convex – this encodes risk aversion on the part of the supervisor.
- 2.4 A is a positively homogeneous cone – scaling a position scales its acceptability requirement proportionally.

Given a reference return  $r$  and an acceptance set  $A$ , the induced risk measure is defined as the smallest amount of cash, invested in the reference instrument, that restores acceptability:

$$\rho_{A,r}(X) = \inf \{ m \in \mathbb{R} \mid m \cdot r + X \in A \} \quad (2.2)$$

Conversely, any risk measure  $\rho$  induces an acceptance set  $A_\rho = \{X \in G \mid \rho(X) \leq 0\}$ . The two constructions are shown (Propositions 2.1–2.2) to be mutual inverses: starting from an acceptance set satisfying Axioms 2.1–2.4 and passing through  $\rho_{A,r}$  back to  $A_\rho$  recovers the closure of the original set, and vice versa. This equivalence is what licenses the rest of the paper to work directly with risk measures rather than acceptance sets.

### 3.2 The Four Coherence Axioms

Translated onto risk measures directly, the axioms on acceptance sets become four properties of  $\rho$ :

$$\text{Axiom } T (\text{translation invariance}): \rho(X + \alpha \cdot r) = \rho(X) - \alpha, \quad \forall X \in G, \alpha \in \mathbb{R} \quad (T)$$

$$\text{Axiom } S (\text{subadditivity}): \rho(X_1 + X_2) \leq \rho(X_1) + \rho(X_2) \quad (S)$$

$$\text{Axiom } PH (\text{positive homogeneity}): \rho(\lambda X) = \lambda \cdot \rho(X), \quad \forall \lambda \geq 0 \quad (PH)$$

$$\text{Axiom } M (\text{monotonicity}): X \leq Y \Rightarrow \rho(Y) \leq \rho(X) \quad (M)$$

A risk measure satisfying all four is called *coherent* (Definition 2.4). A fifth property, Relevance (Axiom R:  $X \leq 0, X \neq 0 \Rightarrow \rho(X) > 0$ ), is discussed separately because it is necessary but not sufficient to rule out undetected risk concentration (Section 3.6 below).

Economically, each axiom does specific work. Translation invariance fixes the units of  $\rho$  to be the same cash units as net worth: adding sure cash  $\alpha$  reduces required capital by exactly  $\alpha$ , so that  $\rho(X + \rho(X) \cdot r) = 0$  – adding exactly the required capital makes the position exactly acceptable. Positive homogeneity models the absence of any liquidity discount for scale: a government does not prevent many firms from independently taking the same position, so doubling a position must double its capital charge (Remark 2 to Axiom PH). Monotonicity is the ordering requirement that a state-by-state dominant position cannot be assessed as riskier; the authors note it specifically rules out the semi-deviation risk measure  $\rho(X) = -E_\mu[X] + \sigma_\mu((X - E_\mu[X])^-)$ , which Axiom S separately excludes, and mean-variance measures  $\rho(X) = -E_\mu[X] + \alpha \cdot \sigma_\mu(X)$ , which Axiom M excludes directly (a position can be shifted to first-order stochastically dominate another while having higher variance, and mean-variance would rank it worse).

Subadditivity carries the paper's central economic argument, restated here in full because it is the axiom the rest of the paper is organized around. ADEH give four independent justifications (Section 2.3, quoted paraphrase): (a) if a measure is not subadditive, an agent wanting to take on risk  $X_1 + X_2$  can split it into two accounts and pay the smaller total  $\rho(X_1) + \rho(X_2)$ , defeating the regulator's intent; (b) a firm facing a non-subadditive capital rule has an incentive to break into separately incorporated affiliates; (c) society rationally requires less capital from a single firm without internal firewalls than from one whose units are legally separated, which is the mirror image of (b); (d) if two trading desks compute  $\rho(X_1)$  and  $\rho(X_2)$  independently and  $\rho$  is subadditive, a supervisor with total capital  $m = m_1 + m_2$  can impose the two limits separately and be assured the

joint position never exceeds  $m$  – true decentralization of a firm-wide constraint into desk-level constraints. Subadditivity is therefore not a technical nicety; it is the property that makes capital allocation compositional.

### 3.3 The Representation Theorem

The paper's central mathematical result gives coherence an exact operational content. Define, for a non-empty family  $P$  of probability measures on  $\Omega$  ("generalized scenarios"),

$$\rho_P(X) = \sup \{ E_P[-X/r] \mid P \in P \} \quad (3.1)$$

Proposition 3.1 shows  $\rho_P$  is always coherent, and satisfies Relevance if and only if the union of the supports of the measures in  $P$  covers  $\Omega$  (every state is "watched" by at least one scenario). Proposition 4.1 – the paper's central theorem – proves the converse: *every* coherent risk measure arises this way.

#### PROPOSITION 4.1 (REPRESENTATION THEOREM)

Given the reference return  $r$ , a risk measure  $\rho$  is coherent if and only if there exists a family  $P$  of probability measures on  $\Omega$  such that  $\rho(X) = \sup \{ E_P[-X/r] \mid P \in P \}$  for all  $X$ .

The "if" direction is immediate: a supremum of linear, monotone functionals is automatically subadditive (the supremum of a sum is at most the sum of suprema), positively homogeneous, translation invariant given  $r$ , and monotone. The "only if" direction is the substantive claim, and its proof strategy is a finite-dimensional separation argument of the kind familiar from the fundamental theorem of asset pricing: because  $\Omega$  is finite,  $G \cong \mathbb{R}^n$ , and a coherent  $\rho$  is a finite, subadditive, positively homogeneous – hence convex, hence continuous – function on  $\mathbb{R}^n$ : a sublinear functional. The classical fact that every sublinear functional on a finite-dimensional space is the support function of a unique closed convex set of linear functionals (an instance of the Hahn–Banach separation theorem, and identified by a referee as mathematically equivalent to Proposition 2.1 in Huber's 1981 *Robust Statistics*) delivers the representing set explicitly:

$$P_\rho = \{ P \in \mathcal{M} \mid \forall Y \in G: E_P[-Y/r] \leq \rho(Y) \} \quad (4.1)$$

Monotonicity and translation invariance are what pin the representing linear functionals down to genuine probability measures rather than arbitrary signed functionals: monotonicity forces non-negative mass on every state (a functional assigning negative weight to some state would be violated by a monotone comparison), and translation invariance anchors total mass to one via normalization against  $r$ . Finiteness of  $\Omega$  is what guarantees the representing measures are countably additive rather than merely finitely additive.

The theorem's immediate corollary is that SPAN – a supremum of expected loss over sixteen prescribed scenarios – is coherent by construction (Proposition 3.1 applied directly to Definition 3.1, the formal version of the SPAN calculation). It also delivers a clean decentralization argument that mirrors point (d) above at the level of scenarios: once a common scenario set is announced, subadditivity is not an assumption but a proven consequence.



### 3.4 Extension from Standard Risks

A dual construction addresses how a partial, pre-agreed pricing or margin rule  $\Psi$  on a restricted family of "standard" risks  $Y$  can be extended into a coherent measure on all of  $G$ . A pair  $(\mu, \gamma)$  of non-negative weights on  $Y$  (finitely many non-zero) and a scalar  $\gamma$  "supports"  $X$  if  $X \geq \sum \mu_Y Y + \gamma \cdot r$ . Given Condition 4.1 – that every support of the zero risk has non-negative cost  $\sum \mu_Y \Psi(Y) - \gamma$  – Proposition 4.2 defines

$$\rho_\Psi(X) = \inf_{(\mu, \gamma)} [\sum \mu_Y \Psi(Y) - \gamma], \text{ the infimum over all supports of } X \quad (4.2)$$

and proves  $\rho_\Psi$  is coherent and is the *largest* coherent measure agreeing with  $\Psi$  on  $Y$  (or majorized by it, in the case of Proposition 4.2's general statement; equality on  $Y$  itself requires the stronger tightness Condition 4.2, delivered in Proposition 4.3). This is the dual mirror image of the scenario representation: whereas adding scenarios to  $P$  makes the measure of Section 3.3 more conservative, having *fewer* standard risks to decompose into makes  $\rho_\Psi$  more conservative – exactly the mechanism, as shown below, by which the SEC/NASD rules overshoot.

### 3.5 Tail Conditional Expectation

Given a quantile-based VaR, define Tail Conditional Expectation as the expected loss conditional on being at or beyond the VaR threshold:

$$TCE_\alpha(X) = -E_P[X/r \mid X/r \leq -VaR_\alpha(X)] \quad (5.1)$$

and Worst Conditional Expectation as the infimum of conditional expected loss over every event of probability exceeding  $\alpha$ :

$$WCE_\alpha(X) = -\inf \{ E_P[X/r \mid A] \mid P[A] > \alpha \} \quad (5.2)$$

WCE is a scenario-based measure in the sense of Definition 3.1 (its scenarios are all conditional measures on sufficiently likely events), hence coherent by Proposition 3.1 directly. Proposition 5.1 shows  $TCE_\alpha \leq WCE_\alpha$  always; Proposition 5.3 shows the two coincide under a uniform base measure with no ties in the distribution of  $X/r$ ; and Proposition 5.4 shows that, under those same regularity conditions, WCE is the smallest law-invariant coherent measure that dominates VaR everywhere – the least conservative coherent repair of VaR obtainable without abandoning coherence. Proposition 5.2 completes the picture from the other direction: VaR itself equals the infimum of *all* coherent measures that dominate it, so VaR is the boundary of the coherent region from below, and TCE/WCE sits at the boundary from just inside.

## 4. Illustrative Method: Institutional Examples as Evidence

Because the paper is axiomatic rather than empirical, its "data" consists of the three institutional risk-measurement rules it sets out to evaluate, plus a battery of constructed counterexamples designed to stress-test VaR. This is a deliberate and defensible substitute for statistical data: the claim under test is a logical one (does a given rule satisfy given axioms?), so a single well-chosen counterexample is dispositive in a way that no amount of historical backtesting could be, since backtesting cannot rule out a rare tail event the sample never contained.

### 4.1 SPAN as a Scenario Measure

SPAN's margin is the worst expected loss across sixteen scenarios – fourteen combinations of a volatility move (up, down, none) and a futures-price move ( $0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1$  of a specified range), plus two "extreme" price moves counted at 35% weight. Recast as Definition 3.1 with fourteen point-mass scenarios and two scenarios that are the convex combination  $(0.35, 0.65)$  of the extreme and no-move outcomes, SPAN is coherent by Proposition 3.1 essentially by inspection.

### 4.2 The SEC/NASD Rules: Coherent but Over-Conservative

The SEC/NASD margin rule decomposes an options portfolio into "spreads"  $S_{H,K} = C_H - C_K$  and charges  $(H-K)^+$  per unit, minimizing total charge over decompositions – a linear program. ADEH work a concrete portfolio: two long calls struck at 10, two short at 20, three short at 30, four long at 40, one short at 50 (a payoff never below  $-10$ ). The SEC method's optimal decomposition requires margin of 30 – three times the true worst-case loss.

Table 1. Illustrative SEC/NASD margin decomposition versus true worst-case loss

| Quantity                                                                          | Value                                |
|-----------------------------------------------------------------------------------|--------------------------------------|
| True worst-case portfolio loss                                                    | 10                                   |
| SEC/NASD margin (optimal spread decomposition)                                    | 30                                   |
| Optimal dual scenario prices $(\pi_{10}, \pi_{20}, \pi_{30}, \pi_{40}, \pi_{50})$ | 10, 10, 10, 0, 0                     |
| Convexity requirement on true call prices                                         | $\pi_{20} + \pi_{40} \geq 2\pi_{30}$ |

Source: reconstructed from Artzner, Delbaen, Eber & Heath (1999), §3.2, Table and linear program (3.1)–(3.4).

The dual of the margin linear program reveals why: its optimal implicit scenario prices satisfy  $\pi_{20} + \pi_{40} = 10 < 2\pi_{30} = 20$ , violating the convexity of call prices in strike that must hold under any arbitrage-free model (Merton, 1973). The rule is therefore implicitly optimizing over scenarios that cannot correspond to any real stock price. In the language of Section 3.4, the SEC/NASD family of "standard risks" (individual spreads only) is too narrow to see that the portfolio can be rebuilt from butterflies  $B_K = C_H - 2C_K + C_H$ , each of which has non-negative payoff by convexity and should require zero margin. **INFERENCE** The diagnosis generalizes beyond options margining: any rule built by decomposing into a restricted family of instruments, rather than by specifying the full scenario



set directly, risks smuggling in implicit scenarios that violate no-arbitrage – a caution equally applicable to modern standardized capital charges (e.g. the Basel standardized approach's risk-weight buckets) that use instrument-level building blocks rather than full scenario revaluation.

### 4.3 Value at Risk: Four Failures of Subadditivity

Value at Risk is defined as the negative of the upper  $\alpha$ -quantile of  $X/r$ :  $VaR_\alpha(X) = -\inf\{x \mid P[X \leq x/r] > \alpha\}$ . It satisfies T, PH, and M, but the paper gives four independent demonstrations that it fails S.

**Table 2. The four VaR subadditivity failures constructed in Section 3.3**

| Example                       | Mechanism                                                                      | Result                                                                                                                                           |
|-------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Digital options (A, B)        | Two 0.8%-probability tail events, each below the 1% VaR threshold individually | Combined 1% VaR is positive; individual VaRs of $-2A$ , $-2B$ are strongly negative – non-convex acceptance set                                  |
| i.i.d. mixed-uniform pair     | Each $X_i$ has 10% downside probability exactly at the 10% quantile boundary   | 10% VaR of $X_1$ and $X_2$ individually is 0; 10% VaR of the sum is strictly positive                                                            |
| Albanese credit concentration | 100 independent 1%-default issuers vs. one concentrated position               | Single-name position: 5% VaR = $-20,000$ ("safe"); diversified position: joint default probability $> 18\%$ , so 5% VaR turns positive ("risky") |
| Pareto-dominated reallocation | Two agents swap tail outcomes across 3 states (probabilities 0.94, 0.03, 0.03) | Reallocated risks Y, Z pass a fixed 5% VaR test with 80 capital each; original X does not, even though $(X, X)$ Pareto-dominates $(Y, Z)$        |

Source: reconstructed from Artzner, Delbaen, Eber & Heath (1999), §3.3, and Albanese (1997), Credit Exposure, Diversification Risk and Coherent VaR, working paper.

The credit-concentration example is the sharpest for a practitioner audience: with a 2% credit spread, 0% base rate, and 1% independent default probability per issuer, a single concentrated 1,000,000 position in one issuer's bonds has 5% VaR equal to  $-20,000$  (the position appears riskless at the 5% level, since default probability 1% is comfortably below the 5% threshold and the spread income is banked). Spreading the same 1,000,000 evenly across 100 independent issuers raises the probability that at least two default simultaneously above 18% – and once diversified, the probability of a negative outcome exceeds the 5% threshold, so 5% VaR turns strictly positive. EVIDENCE VaR here does not merely misprice risk at the margin; it inverts the sign of the risk assessment between the concentrated and diversified position, rewarding exactly the exposure a regulator should most want to discourage.

The paper also notes the one regime where VaR behaves well: under joint normality,  $VaR_\alpha(X) = -(E_p[X] + \Phi^{-1}(\alpha) \cdot \sigma_p(X))$ , and because  $\sigma_{X+Y} \leq \sigma_X + \sigma_Y$  always holds, VaR is subadditive at any confidence level above 50%. The failures are therefore not a mathematical curiosity but a direct consequence of the non-normal features – skew, fat tails, discontinuous (digital) payoffs – that characterize exactly the positions (credit portfolios, options books) for which risk measurement matters most.

## 5. Results

### 5.1 The Coherence Classification of Existing Methods

Table 3 summarizes the paper's central classification result, which functions as its main "finding" in the sense of an empirical paper's headline coefficient.

**Table 3.** Coherence status of the three examined risk-measurement methods

| Method                               | T | S | PH | M | Coherent?                                                               |
|--------------------------------------|---|---|----|---|-------------------------------------------------------------------------|
| SPAN (CME)                           | ✓ | ✓ | ✓  | ✓ | Yes – direct instance of scenario representation                        |
| SEC/NASD margin rules                | ✓ | ✓ | ✓  | ✓ | Yes, but strictly dominated by a less conservative coherent alternative |
| Value at Risk                        | ✓ | ✗ | ✓  | ✓ | No – four independent counterexamples to subadditivity                  |
| Tail / Worst Conditional Expectation | ✓ | ✓ | ✓  | ✓ | Yes – scenario measure over tail-conditional distributions              |

Source: author's synthesis of Propositions 3.1, and Sections 3.2, 3.3, 5.1 of Artzner, Delbaen, Eber & Heath (1999).

Economically, the magnitude in the SEC/NASD case (margin of 30 against a true worst case of 10, a 3x overcharge on this particular portfolio) is a statement about capital efficiency: coherent but non-minimal rules tie up capital that could otherwise be deployed, a real cost to the regulated firm even though the rule never under-protects the regulator. The VaR failures are qualitatively different in kind: they are not inefficiencies but sign errors, in which a genuinely more dangerous position (the diversified credit book, the combined digital-option book) receives a strictly smaller risk charge than a genuinely safer one.

### 5.2 Extension and Consistency with Arbitrage-Free Pricing

Section 4.3 of the paper (reconstructed here) links the scenario representation to the absence of arbitrage. Let  $Q^r$  denote the set of pricing measures consistent with an arbitrage-free market using  $r$  as numeraire, and  $P_\rho$  the scenario set representing a coherent measure  $\rho$ . If these two convex sets intersect (Condition 4.3), every accepted position is priced non-negatively by at least one pricing measure the risk measure also regards as relevant, and risk cannot accumulate costlessly. If they are disjoint, the separation theorem produces a position  $Z$  that is exactly on the boundary of acceptability ( $\rho(Z) = 0$ ) yet strictly negatively priced under every arbitrage-free pricing measure – an accepted position that the market itself treats as a net liability. This result operationalizes the caution attached to the Relevance axiom (Section 3.2): relevance alone does not prevent undetected concentration; the scenario set used for capital purposes must actually overlap the market's own pricing measures.

## 6. Robustness and Additional Results

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Because the paper's claims are theorems rather than estimated coefficients, "robustness" here means the extent to which the representation and extension results hold under weaker hypotheses, and whether the proposed remedy (TCE) is tight.

### 6.1 Tightness of the Extension Construction

Proposition 4.2's extension  $\rho_\Psi$  is proved to be the *largest* coherent measure majorized by an arbitrary consistent  $\Psi$  on the standard risks – a general result that holds regardless of which family of standard risks is chosen. Proposition 4.3 sharpens this to exact equality on the standard-risk family itself, but only under the stronger tightness Condition 4.2 (every support of a standard risk already costs at least as much as  $\Psi$  assigns to it directly). **CRITIQUE** This conditionality is a real limitation for practical use: a regulator adopting the extension methodology must verify Condition 4.2 case by case, and the paper does not supply a general, easily checked sufficient condition on  $\Psi$  beyond the definition itself – the SEC/NASD case is solved by hand, not by a generic algorithm.

### 6.2 The Combination of Independent Risk Measures

Proposition 5.5 (reconstructed) shows that when a firm has separate coherent measures  $\rho_1, \rho_2$  for, say, credit risk and market risk on independent state spaces  $\Omega_1, \Omega_2$ , the largest coherent extension combining both is exactly the scenario measure generated by all joint distributions on  $\Omega_1 \times \Omega_2$  whose marginals lie in the two separate scenario sets  $P_1, P_2$ . This is a robustness result of a different sort: it shows the framework is closed under aggregation across risk types built by different internal models – a property neither SPAN nor the SEC/NASD approach, taken piecemeal, is shown to have.

### 6.3 How Close Is TCE to VaR?

Propositions 5.1–5.4 together bound how conservative the proposed remedy needs to be. TCE is always at most WCE (Proposition 5.1); the two coincide under a uniform base measure with no ties (Proposition 5.3); and under those conditions WCE is provably the *smallest* law-invariant coherent measure dominating VaR (Proposition 5.4), while VaR itself is the infimum over all coherent measures dominating it (Proposition 5.2). Read together, these results say TCE/WCE is not an arbitrarily conservative fix chosen for convenience – it sits as close to VaR as coherence permits, among measures built from the same distributional information. **CRITIQUE** The uniform-measure and no-tie assumptions in Proposition 5.4 are restrictive; under a genuinely continuous, non-uniform distribution (the empirically relevant case for returns), the paper's proof does not directly establish that WCE remains the tightest coherent dominator, only that TCE and WCE coincide (Proposition 5.3 requires uniformity specifically for the equality, not merely the coherence).

## 7. Critical Evaluation

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### 7.1 What the Paper Establishes, Conditionally, and Only Suggests

#### ESTABLISHED

Coherence (T, S, PH, M) is logically equivalent to the scenario-representation formula (3.1); VaR fails subadditivity, demonstrated by four independent, fully worked counterexamples rather than by simulation; SPAN is coherent by direct construction; the SEC/NASD rule is coherent but strictly capital-inefficient on the worked example; TCE and WCE are coherent.

#### SUPPORTED BUT CONDITIONAL

TCE and WCE coincide, and WCE is the tightest law-invariant coherent measure dominating VaR, but only under the stated regularity conditions (uniform base probability, no ties in the loss distribution) that do not hold for continuous asset-return distributions in general; the extension of a partial rule  $\Psi$  to a coherent measure equal to  $\Psi$  on the standard risks (rather than merely majorized by it) requires the case-specific tightness Condition 4.2.

#### SUGGESTIVE

That TCE/WCE, rather than some other point in the coherent class, is the "right" practical choice for regulators is argued from proximity to existing VaR-based practice and from the dominance results above, not from a demonstration that TCE uniquely optimizes any independently motivated loss function (e.g. a specific utility or regulatory objective).

#### NOT ESTABLISHED

The paper does not establish that coherence, on its own, is sufficient for a risk measure to be usable in practice: it says nothing about estimability, statistical stability of the tail expectation under limited data, or whether a coherent measure can be reliably backtested – concerns that would only be resolved (and partly reopened) by the later elicibility literature (Section 9 below).

### 7.2 Strengths

**PAPER** The authors themselves emphasize the generality of the representation theorem (it subsumes SPAN, generalizes cleanly to multi-currency and multi-risk-type settings, and requires no distributional assumption) and the fact that the axioms are independently motivated by operational concerns (decentralization, mergers, regulatory arbitrage) rather than by mathematical convenience. **CRITIQUE** Beyond the authors' own claims, three further strengths stand out on independent reading. First, the argument is constructive rather than merely negative: rather than only proving VaR inadequate, the paper derives its replacement from the same axioms, so the critique and the remedy come from one unified apparatus. Second, the counterexamples in Section 3.3 are deliberately chosen to be economically realistic (credit portfolios, digital options,

small discrete state spaces) rather than pathological edge cases invented purely to break a formula – the credit-concentration example in particular has an obvious real-world analogue in structured-credit portfolios. Third, the paper's insistence on working without a complete-markets assumption is unusual for 1996-vintage mathematical finance and is precisely what makes the framework applicable to non-market (e.g. insurance, credit) risks as well as market risk, a scope later realized in Basel's treatment of credit and operational risk capital.

### 7.3 Weaknesses and Independent Critique

The most important weakness, independent of the limitations authors already flag, is that **coherence does not imply statistical estimability**. A supremum over a rich scenario set, or a conditional tail expectation, is intrinsically harder to estimate from a finite historical sample than a quantile: VaR estimation requires only ordering observations, while TCE requires averaging the (typically sparse) tail observations beyond the quantile, and WCE requires an optimization over conditioning events that is not spelled out algorithmically for continuous distributions. The paper is silent on sampling error, and this silence became consequential fifteen years later when Gneiting (2011) proved Expected Shortfall is not elicitable – there is no scoring function whose expected value is minimized by the true ES, unlike VaR, which is a quantile and is elicitable by construction. **CRITIQUE** This is a genuine tension the paper does not anticipate: the property that makes VaR axiomatically deficient (it only sees a threshold, not the tail beyond it) is closely related to the property that makes it statistically well-behaved (it is a simple order statistic), and the property that makes TCE axiomatically superior is related to what makes it harder to backtest reliably in real time.

A second weakness concerns the single-period, finite-state setting. The entire representation theorem rests on  $G \cong \mathbb{R}^n$  being finite-dimensional; the authors flag in a remark that extending to infinite  $\Omega$  would in general only recover finitely additive (not countably additive) scenario measures, a much weaker and less tractable object. The multi-period extension is explicitly deferred ("our current results in the simplified setting represent a first step," Section 2.1), and the dynamic, time-consistent version of coherent (and later convex) risk measures required substantial further work by other authors (see Section 9).

A third, more conceptual, weakness: the four axioms characterize a large class, not a point, and the paper is candid about this ("our axioms are not restrictive enough to specify a unique risk measure," Introduction, point 11). **CRITIQUE** This is a virtue for generality but a cost for a working regulator, who must still choose among the coherent class using criteria the paper does not supply. The choice of TCE over, say, a more elaborate spectral risk measure (weighting all tail quantiles rather than a single conditional average) is defended by regulatory proximity, not derived from the axioms.

The single most important weakness, weighing the above three, is the estimability gap: an axiomatic property that cannot be reliably measured in finite samples has limited value as a regulatory standard until a workable estimation and backtesting procedure exists – a gap the paper leaves entirely open and that took over a decade of subsequent literature (Acerbi and Tasche, 2002; Gneiting, 2011; Fissler and Ziegel, 2016) to close, and even then only partially.

## 8. Contradictions and Edge Cases

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Pressing the framework at its boundaries clarifies what coherence does and does not guarantee.

*Does the mechanism fail at extreme values?* Consider the degenerate case  $P = \{P_1, P_2\}$  with two scenarios that are mutually singular (disjoint supports). The resulting  $\rho_p$  is coherent by Proposition 3.1, but it will assign the same worst-case risk to a position concentrated entirely in the support of  $P_1$  as to one that is a probability-weighted mixture spanning both supports, provided the worst expectation is achieved on the same scenario – illustrating that coherence, alone, does not penalize concentration within a single scenario's support any more finely than the scenario itself resolves states. This is exactly the gap the Relevance axiom and the Section 4.3 pricing-consistency condition are needed to close, and even together they only guarantee that *some* pricing-consistent scenario notices the concentration, not that the chosen scenario set does.

*What happens under extreme positive homogeneity?* Axiom PH implies a position ten times the size of a benchmark position costs exactly ten times the capital, with no allowance for the fact that liquidating a position ten times larger may take proportionally longer and move the market against the liquidator. The authors note this explicitly (Remark 1 to Axiom PH) as a modeling choice rather than a universal truth, and flag that if liquidity effects are material, the "future net worth" itself should already impound the cost of an orderly unwind – pushing the problem into the definition of  $X$  rather than into the axioms on  $\rho$ . In practice, large or illiquid positions (a bank's own thinly traded structured book) are exactly where this linearity assumption is least credible, which is the intellectual seed of the later liquidity-adjusted VaR and liquidity-augmented coherent risk measure literature.

*Could another mechanism generate the same VaR failures?* One might ask whether the digital-option and credit-concentration examples are artifacts of using VaR specifically, rather than genuine failures of quantile-based thinking as such. They are not artifacts: any measure defined purely as a threshold exceedance probability, rather than an expectation beyond the threshold, will exhibit the same qualitative failure, because the mechanism in every one of the four examples is that VaR is blind to what happens beyond the quantile it reports. Any statistic sharing that blindness – not VaR specifically, but the entire class of pure quantile risk statistics – inherits the same subadditivity failure. This generalization is implicit in the representation theorem: since no pure quantile can be written as a supremum of expectations (a quantile is not sublinear in general), no member of that class can be coherent, a fact the paper does not state in these terms but which follows directly from Proposition 4.1.

*Does the result survive outside a single regulatory jurisdiction?* The multi-currency remark in Section 2.3 shows acceptability is currency-invariant when acceptance sets across currencies are linked by the exchange rate ( $e_i A_i = A_1$  for all  $i$ ), but the induced capital requirement  $\rho_{A_i, r_i}(X)$  in one country's reference instrument need not equal the requirement  $\rho_{A_j, r_j}(X)$  a second regulator would compute in its own reference instrument, even at a common initial exchange rate of one. **INFERENCE** This means coherence, as defined, is not by itself sufficient to guarantee consistent capital treatment of



the same global position across two coherent national regulators – an edge case with direct relevance to the cross-border consistency debates that later accompanied Basel III implementation.

## 9. Further Literature

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### 9.1 Preceding and Contemporaneous Work

**Merton (1973)**, *Theory of Rational Option Pricing* – supplies the convexity-of-call-prices-in-strike result that exposes the SEC/NASD rule's implicit-scenario failure in Section 4.2. **Huber (1981)**, *Robust Statistics* – the sublinear-functional representation theorem ADEH's referee identified as mathematically equivalent to the paper's own Proposition 4.1, situating the result within a pre-existing statistical tradition rather than as an isolated finance-specific discovery. **RiskMetrics Technical Document (Morgan Guaranty, 1995)** and **Duffie and Pan (1997)** – the practitioner and academic statements of the VaR methodology the paper sets out to critique. **Albanese (1997)** – the credit-concentration example (Table 2, row 3) that the paper adopts directly, connecting the critique to credit portfolio management, a field distinct from the market-risk VaR literature that originated the statistic.

### 9.2 Direct Descendants

**Föllmer and Schied (2002)** and, independently, **Frittelli and Rosazza Gianin (2002)**, generalize coherence to *convex* risk measures by relaxing positive homogeneity and subadditivity to joint convexity alone, allowing risk measures that penalize scale increasingly (a liquidity-sensitive relaxation of exactly the PH concern raised in Section 8 above); their representation theorem replaces the scenario supremum with a penalized supremum,  $\rho(X) = \sup_Q \{E_Q[-X] - \alpha(Q)\}$ . **Acerbi and Tasche (2002)**, "Expected Shortfall: A Natural Coherent Alternative to Value at Risk," repair a gap in ADEH's own TCE/WCE construction: TCE as defined in equation (5.1) is not coherent in general for distributions with atoms (discrete or mixed distributions), and Acerbi and Tasche supply a corrected integral definition of Expected Shortfall,  $ES_\alpha(X) = \alpha^{-1} \int_0^\alpha VaR_u(X) du$ , that is coherent for every distribution, continuous or not – the version that ultimately entered regulation. **Rockafellar and Uryasev (2000)**, "Optimization of Conditional Value-at-Risk," show CVaR (equivalent to WCE/ES) can be minimized by linear or convex programming without first computing VaR, making the coherent measure practically optimizable in portfolio construction, not merely computable after the fact. **Acerbi (2002)** generalizes further to spectral risk measures, weighting the entire loss distribution by an arbitrary risk-aversion function rather than a single tail cutoff, of which VaR, TCE, and WCE are special or limiting cases. **Kusuoka (2001)** gives the definitive representation of law-invariant coherent risk measures as mixtures of Expected Shortfall across confidence levels, closing the question of what the "natural" coherent measures look like once one restricts to measures depending only on distribution (not state identity).

### 9.3 The Regulatory and Statistical Aftermath

The Basel Committee's **Fundamental Review of the Trading Book** (finalized 2016, phased in through the early 2020s) replaced 99% VaR with 97.5% Expected Shortfall as the standard market-risk capital metric, citing subadditivity and tail-sensitivity concerns that trace directly to this paper's Section 3.3 – the single clearest instance of this literature's direct legislative adoption. That adoption, however, reopened exactly the estimability question flagged in Section 7.3 above: **Gneiting (2011)** proved Expected Shortfall is not elicitable (no scoring function's expected value is minimized by the true ES), which was widely read as implying ES could not be properly backtested, a claim that generated substantial controversy until **Fissler and Ziegel (2016)** proved ES is jointly elicitable together with VaR, restoring a workable (if more complex) backtesting methodology. This entire thirty-year arc – axiomatic critique, regulatory adoption, statistical counter-critique, statistical repair – has this paper as its unambiguous starting point.

## 10. Research Gaps

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**Paper-level gap.** The single-period, finite-state model does not address time-consistency: how should a risk assessment made today relate to a re-assessment tomorrow as information arrives? The paper explicitly defers this ("a first step," Section 2.1). *Why it matters:* real risk management is continuous, and a measure that is coherent at each date but time-inconsistent across dates can be gamed by an agent who re-optimizes at each rebalancing. *Possible way forward:* dynamic and conditional risk measures (surveyed later by Föllmer and Schied's dynamic extensions), imposing an additional recursive consistency axiom across the filtration.

**Paper-level gap.** Estimability and backtesting of scenario-based or tail-conditional measures from finite samples is never addressed. *Why it matters:* a regulatory standard that cannot be verified against realized data is not enforceable in the way VaR, however flawed, could be checked by counting quantile exceedances. *Possible way forward:* the elicibility and joint-elicibility literature (Gneiting 2011; Fissler and Ziegel 2016) directly targets this, though it arrived over a decade later and as a reaction to, not an anticipation by, this paper.

**Literature-level gap.** No canonical method exists, even after Kusuoka's (2001) characterization, for choosing a specific coherent measure (or specific confidence level and weighting function) on economic rather than proximity-to-precedent grounds. *Why it matters:* the choice of confidence level (97.5% ES versus 99% VaR, as in FRTB) has first-order effects on measured capital, yet is set by regulatory negotiation rather than derived from a stated loss function of the regulator. *Possible way forward:* tying the choice of risk-aversion weighting in a spectral risk measure to an explicit macroprudential objective function (e.g. minimizing the expected fiscal cost of bank failure) rather than treating the weighting as a free parameter.

**Methodological gap.** The extension theorem (Section 3.4) requires case-by-case verification of Condition 4.2 and offers no general algorithm for finding the tightest coherent extension of an arbitrary partial rule across many instrument classes simultaneously (equities, credit, rates, commodities). *Why it matters:* exactly this multi-asset-class aggregation problem is what modern

bank-wide economic capital models must solve. *Possible way forward:* computational approaches (large-scale linear or convex programming) that search over supporting decompositions automatically, rather than by hand as in the paper's worked example.

## 11. Research Opportunities

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### 1. Elicitability-Constrained Coherent Risk Measures for Credit Portfolios

**RESEARCH QUESTION** Among coherent risk measures that are also (jointly) elicitable, which minimizes estimation variance for realistic credit-loss distributions with the concentration structure of ADEH's Albanese example?

**MOTIVATION** ADEH show VaR fails exactly where credit concentration matters most, but the elicibility literature that repairs ES's backtestability was built for market-risk (roughly continuous, thin-tailed-relative-to-credit) returns, not for the discrete, highly skewed loss distributions typical of credit portfolios.

**HYPOTHESIS** Standard Expected Shortfall at conventional confidence levels (97.5–99%) is estimated with substantially higher variance on credit-loss data than on market-return data, because credit losses are rare and lumpy, undermining the practical case for ES in credit capital models specifically.

**ECONOMIC MECHANISM** Credit losses are driven by discrete default events with low, correlated probabilities; a small sample of realized defaults gives a noisy estimate of the tail expectation even when the model is correctly specified.

**LITERATURE CONNECTION** Extends Acerbi and Tasche (2002) and Fissler and Ziegel (2016) from market to credit risk; directly tests whether ADEH's own worked example (Table 2, row 3) is estimation-robust.

**DATA** Moody's or S&P historical corporate default and recovery databases; simulated portfolios of varying concentration to control ground truth.

**VARIABLES** Realized default indicator per issuer-year, loss-given-default, portfolio concentration (effective number of names), confidence level.

**METHOD** Backtesting simulation study comparing estimator variance of VaR, ES, and a spectral measure across confidence levels and portfolio concentrations, using the Fissler-Ziegel joint scoring function.

**IDENTIFICATION STRATEGY** Not causal; a Monte Carlo and historical-backtest comparison of estimator properties under a controlled data-generating process.

**EXPECTED CONTRIBUTION** A quantitative case for (or against) applying market-risk-style coherent measures unmodified to credit portfolios, with a proposed concentration-adjusted confidence level if needed.

**MAIN RISK** Historical default data are sparse for exactly the tail events that matter, limiting statistical power regardless of estimator choice.

## 2. A Liquidity-Augmented Coherent Risk Measure

**RESEARCH QUESTION** Can positive homogeneity (Axiom PH) be relaxed to a liquidity-dependent sublinear (rather than linear) scaling rule while preserving a clean representation theorem analogous to Proposition 4.1?

**MOTIVATION** Section 8 above identifies PH's linearity-in-size assumption as counterfactual for large or illiquid positions; ADEH themselves flag this as an open modeling choice rather than resolve it.

**HYPOTHESIS** Replacing PH with a concave (sub-linear-in-scale) requirement on  $\rho(\lambda X)/\lambda$  yields a representation as a scenario supremum weighted by a size-dependent liquidation cost function, nesting Föllmer-Schied convexity as a special case.

**ECONOMIC MECHANISM** Large positions cannot be liquidated instantaneously at the quoted price; the true "future net worth" of a large position embeds a market-impact cost increasing in size.

**LITERATURE CONNECTION** Builds on Föllmer and Schied (2002) convex risk measures and the market-impact literature (Almgren-Chriss-style execution cost models), bridging execution microstructure with the ADEH axiomatic program.

**DATA** Intraday order-book data for large-cap equities and corporate bonds to calibrate a market-impact function.

**VARIABLES** Position size relative to average daily volume, bid-ask spread, realized impact cost, liquidation horizon.

**METHOD** Derive the representation theorem under the relaxed axiom set; calibrate the liquidation-cost function empirically and compare resulting capital charges to standard coherent measures on real trading books.

**IDENTIFICATION STRATEGY** Theoretical derivation plus calibration; not a causal empirical design.

**EXPECTED CONTRIBUTION** A formally coherent (in a generalized sense) risk measure suitable for large or concentrated books, directly answering the open question ADEH pose but do not resolve.

**MAIN RISK** The generalized axioms may fail to deliver a representation theorem as clean as Proposition 4.1, weakening the theoretical payoff relative to the empirical calibration effort required.

### 3. Cross-Jurisdictional Consistency of Coherent Capital Requirements

**RESEARCH QUESTION** For a multinational bank subject to coherent capital rules in several jurisdictions, how large in practice is the wedge between  $\rho_{A_i, r_i}(X)$  and  $\rho_{A_j, r_j}(X)$  that the paper's own multi-currency remark (Section 2.3) shows can exist even when acceptance sets are currency-linked?

**MOTIVATION** Basel implementation varies by jurisdiction in confidence level, model approval, and reference instrument conventions; ADEH's own theory predicts this wedge should exist even under otherwise identical coherent rules, but its empirical size is unmeasured.

**HYPOTHESIS** The wedge is economically material (exceeding typical bank capital buffers) for portfolios with material exposure to currencies whose reference-rate volatility is high relative to the position's own risk.

**ECONOMIC MECHANISM** Each jurisdiction's regulator effectively applies its own scenario set (via Proposition 4.1) to a currency-converted version of the same net worth; different reference instruments and scenario sets need not commute with the currency conversion.

**LITERATURE CONNECTION** Extends ADEH's own Remark on multi-currency acceptance sets (Section 2.3) into an empirical/regulatory-economics research question; connects to the Basel cross-border consistency literature.

**DATA** Regulatory capital disclosures (Pillar 3 reports) of globally systemically important banks across jurisdictions; exchange rate volatility data.

**VARIABLES** Reported capital requirement by jurisdiction and business line, currency composition of trading book, realized exchange-rate volatility.

**METHOD** Comparative panel analysis of disclosed capital ratios across jurisdictions for banks with comparable, currency-decomposed trading books; a simulation exercise applying ADEH's own formal multi-currency framework to stylized portfolios.

**IDENTIFICATION STRATEGY** Cross-sectional comparison controlling for portfolio composition; primarily descriptive/theoretical rather than causal.

**EXPECTED CONTRIBUTION** First empirical quantification of a theoretical gap ADEH identify but do not measure, relevant to ongoing Basel III finalization debates on cross-border consistency.

**MAIN RISK** Regulatory capital disclosures may not be granular enough to isolate the currency-conversion wedge from other sources of jurisdictional divergence (model approval status, add-ons).

#### 4. Replication and Extension of the SEC/NASD Overshoot Result to Modern Standardized Approaches

**RESEARCH QUESTION** Does the Basel standardized approach for market risk (the "sensitivities-based method" under FRTB) exhibit the same implicit-scenario, convexity-violating overshoot the paper finds in the 1990s SEC/NASD rule?

**MOTIVATION** The SEC/NASD diagnosis (Section 4.2) generalizes to any bucketed, instrument-decomposition-based standardized capital rule; the current FRTB standardized approach is structurally similar (risk-weight buckets, prescribed correlations) and has not been checked against this specific failure mode.

**HYPOTHESIS** There exist realistic options or credit-derivative portfolios for which the FRTB standardized capital charge exceeds the true worst-case loss by a similar multiple to the 3x overshoot ADEH find for the SEC rule.

**ECONOMIC MECHANISM** Standardized risk-weight buckets are, in ADEH's terms, a restricted family of "standard risks"  $Y$ ; if the family cannot express certain risk-reducing combinations (butterflies, cross-bucket hedges), the extension of Section 3.4 is forced to be more conservative than necessary.

**LITERATURE CONNECTION** Direct methodological replication of ADEH Section 3.2's linear-programming argument, applied to a live regulatory rule three decades later.

**DATA** Published FRTB standardized-approach risk-weight and correlation tables (Basel Committee texts); constructed test portfolios.

**VARIABLES** Standardized capital charge, true worst-case loss under full revaluation, portfolio Greeks/sensitivities.

**METHOD** Construct the analogous linear program (as in equations 3.1–3.4 above) for representative derivatives books under the FRTB bucket structure; solve for worst-case overshoot portfolios.

**IDENTIFICATION STRATEGY** Not causal; a constructive/existence-proof exercise mirroring the original paper's method exactly.

**EXPECTED CONTRIBUTION** A direct, methodologically faithful update of one of the paper's two central worked examples to current regulation, of practical interest to bank treasury and regulatory-capital functions.

**MAIN RISK** Modern standardized approaches include more granular buckets and hedging recognition than the 1990s SEC/NASD rule, which may have already closed much of the gap the original paper found.



## 5. Time-Consistent Dynamic Extension of Tail Conditional Expectation

**RESEARCH QUESTION** Can a dynamic, multi-period version of TCE/WCE be constructed that is both coherent at each date and time-consistent (a position acceptable today given tomorrow's acceptable continuations is itself acceptable), closing the gap ADEH explicitly leave open?

**MOTIVATION** ADEH's single-period model is silent on dynamic consistency; the subsequent dynamic risk measure literature developed largely independently of, rather than as a direct extension of, ADEH's specific TCE proposal.

**HYPOTHESIS** A recursively defined dynamic TCE, updated via conditional scenario sets at each date, satisfies time consistency only under an additional restriction on how the scenario family evolves (analogous to a rectangularity condition in robust control).

**ECONOMIC MECHANISM** Time consistency requires that risk assessments made with more information (later dates) be compatible with assessments made with less information (earlier dates); naively updating a static scenario-based measure period-by-period generally breaks this compatibility unless the scenario sets are constructed recursively.

**LITERATURE CONNECTION** Directly extends ADEH's Section 2.1 remark that the single-period model is "a first step"; connects to Föllmer-Schied dynamic convex risk measures and the rectangularity conditions in robust dynamic programming.

**DATA** Not primarily empirical; a theoretical derivation, illustrated with a simulated multi-period trading book.

**VARIABLES** Not applicable in the primary theoretical component; simulated portfolio value paths for the illustration.

**METHOD** Formal derivation of recursive TCE under a rectangular scenario-updating rule; numerical illustration on a simulated multi-period book comparing naive period-by-period TCE to the time-consistent version.

**IDENTIFICATION STRATEGY** Not applicable; a theoretical/constructive contribution.

**EXPECTED CONTRIBUTION** A direct, named completion of the specific open problem ADEH pose, using their own preferred measure (TCE) rather than a generic convex risk measure.

**MAIN RISK** The dynamic risk-measure literature is already extensive; the specific TCE-based recursive construction may turn out to be a special case of results already established for general coherent or convex measures, limiting novelty.

## 11.1 Ranking

Ranked by a combination of novelty, feasibility, data availability, methodological quality, and economic importance, the top three ideas are, in order: **Idea 4** (replicating the SEC/NASD overshoot result against live FRTB rules) ranks highest on feasibility and economic importance – it requires only public regulatory texts and constructed portfolios, directly informs current bank regulatory-capital practice, and is a faithful methodological extension rather than a speculative one. **Idea 1** (elicitable-constrained credit risk measures) ranks second: it is highly novel, addresses a real and current tension in the literature (the elicitable debate has not been resolved for credit portfolios specifically), and is feasible with existing default databases, though data sparsity in the tail is a genuine limitation. **Idea 5** (time-consistent dynamic TCE) ranks third: it is the most conceptually important gap ADEH themselves flag, but its novelty is tempered by an

already substantial dynamic risk-measure literature, and its primarily theoretical nature makes the expected contribution harder to make concrete without a strong applied illustration. Ideas 2 and 3 are promising but rank lower here primarily on feasibility grounds – idea 2 requires a new representation theorem with uncertain existence, and idea 3 requires regulatory disclosure data that may not be granular enough to isolate the effect of interest.

## 12. Learning Framework

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### 12.1 Prerequisite Concepts

Convex analysis on finite-dimensional spaces (sublinear functionals, support functions, separation theorems); basic measure theory on finite state spaces; the definition and basic properties of Value at Risk and quantiles; the no-arbitrage principle and convexity of option prices in strike (Merton, 1973); linear programming duality.

### 12.2 Core Concepts Learned

The distinction between an acceptance set and the risk measure it induces, and their formal equivalence; the four coherence axioms and the specific economic failure each rules out; the representation of coherent measures as worst-case expectations over generalized scenarios; the mechanism by which VaR fails subadditivity; the construction of a coherent measure from a partial rule on standard risks via the extension theorem; Tail Conditional Expectation and Worst Conditional Expectation as coherent, VaR-dominating measures.

### 12.3 Important Equations

The coherence axioms (T, S, PH, M); the risk-measure/acceptance-set correspondence (2.2)–(2.3); the scenario representation (3.1) and its converse, the representation theorem (4.1); the extension formula (4.2); the definitions of VaR, TCE, and WCE (5.1)–(5.2).

### 12.4 Methods Learned

Constructing counterexamples to test an axiom (subadditivity) rather than merely estimating a statistic; using linear programming duality to reveal an implicit, economically inconsistent scenario set hidden inside a decomposition-based rule; applying finite-dimensional separation theorems to obtain a dual representation of a convex functional; distinguishing law-invariant from state-dependent properties of a risk measure.

### 12.5 Questions to Test Understanding

**Comprehension.** State the four coherence axioms and explain, in one sentence each, the practical failure each is designed to prevent. Explain why VaR satisfies T, PH, and M but not S. **Analysis.** Explain why the representation theorem's "only if" direction requires  $\Omega$  to be finite, and what is lost (or requires additional structure) if  $\Omega$  is infinite. Explain, using the LP duality argument of

Section 4.2, exactly why the SEC/NASD rule overcharges the worked example by a factor of three. **Research.** Propose a modification of Axiom PH that would make the framework sensitive to liquidation cost, and identify which of Propositions 2.1–5.5 would need to be re-proved under your modification. Assess whether Kusuoka's (2001) characterization of law-invariant coherent measures as mixtures of Expected Shortfall renders TCE/WCE's specific proposal in this paper redundant, partially redundant, or still independently useful.

## 12.6 Papers to Read Next

Föllmer and Schied (2002), "Convex Measures of Risk and Trading Constraints"; Acerbi and Tasche (2002), "Expected Shortfall: A Natural Coherent Alternative to Value at Risk"; Rockafellar and Uryasev (2000), "Optimization of Conditional Value-at-Risk"; Kusuoka (2001), "On Law Invariant Coherent Risk Measures"; Gneiting (2011), "Making and Evaluating Point Forecasts."

## 13. Conclusion

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### One-Sentence Thesis

A risk measure is defensible for capital-requirement purposes only if it is coherent – translation invariant, subadditive, positively homogeneous, and monotone – and coherence is exactly equivalent to reporting the worst-case expected loss across an explicit set of scenarios, a characterization that vindicates SPAN, exposes the SEC/NASD rule as needlessly conservative, and proves Value at Risk structurally unsound.

### Three Most Important Insights

First, subadditivity is not a mathematical refinement but the property that makes decentralized risk limits and firm-wide capital add up correctly; without it, capital allocation across desks or legal entities becomes internally inconsistent. Second, every coherent risk measure, without exception, is a worst-case scenario calculation in disguise – there is no coherent alternative to "define your scenarios and take the supremum." Third, VaR's failure is not a modeling detail correctable by better calibration; it is structural, demonstrated by exact counterexamples, and specifically manifests where diversification and tail risk interact, which is precisely where risk measurement matters most.

### Strongest Evidence

The Albanese credit-concentration example: a single-name position judged riskless (negative 5% VaR) while the fully diversified version of the same exposure is judged risky (positive 5% VaR) is a sign-reversal, not a mere inefficiency, and it is fully worked with explicit numbers rather than asserted qualitatively.

## Biggest Weakness

The paper offers no treatment of statistical estimability: it is silent on how reliably a scenario-based or tail-conditional measure can be estimated from finite historical data, a silence that became a live regulatory problem once Expected Shortfall replaced VaR under FRTB and Gneiting's (2011) non-elicitability result reopened the backtesting question the paper never posed.

## Most Important Research Gap

The absence of a time-consistent, multi-period version of the framework: the entire representation theorem is single-period and finite-state, and the paper explicitly defers the dynamic extension as future work that it does not itself undertake.

## Best Research Opportunity

Testing whether the SEC/NASD overshoot mechanism of Section 4.2 – a restricted family of standard risks implicitly encoding an arbitrage-inconsistent scenario set – recurs in the FRTB standardized approach that is now live regulation, using the same linear-programming method the original paper used three decades ago.

## Three Papers to Read Next

Föllmer and Schied (2002), "Convex Measures of Risk and Trading Constraints," *Finance and Stochastics*; Acerbi and Tasche (2002), "Expected Shortfall: A Natural Coherent Alternative to Value at Risk," *Economic Notes*; Gneiting (2011), "Making and Evaluating Point Forecasts," *Journal of the American Statistical Association*.

## 14. References

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- Acerbi, C. (2002). Spectral measures of risk: a coherent representation of subjective risk aversion. *Journal of Banking & Finance*, 26(7), 1505–1518.
- Acerbi, C., and Tasche, D. (2002). Expected shortfall: a natural coherent alternative to value at risk. *Economic Notes*, 31(2), 379–388.
- Albanese, C. (1997). *Credit Exposure, Diversification Risk and Coherent VaR*. Working paper, Department of Mathematics, University of Toronto.
- Artzner, P., Delbaen, F., Eber, J.-M., and Heath, D. (1997). Thinking coherently. *Risk*, 10(11), 68–71.
- Artzner, P., Delbaen, F., Eber, J.-M., and Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3), 203–228.
- Basel Committee on Banking Supervision (1996). *Amendment to the Capital Accord to Incorporate Market Risks*. Bank for International Settlements, Basel.
- Basel Committee on Banking Supervision (2019). *Minimum Capital Requirements for Market Risk*. Bank for International Settlements, Basel.

- Duffie, D., and Pan, J. (1997). An overview of value at risk. *Journal of Derivatives*, 4(3), 7–49.
- Fissler, T., and Ziegel, J. F. (2016). Higher order elicibility and Osband's principle. *Annals of Statistics*, 44(4), 1680–1707.
- Föllmer, H., and Schied, A. (2002). Convex measures of risk and trading constraints. *Finance and Stochastics*, 6(4), 429–447.
- Frittelli, M., and Rosazza Gianin, E. (2002). Putting order in risk measures. *Journal of Banking & Finance*, 26(7), 1473–1486.
- Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494), 746–762.
- Huber, P. J. (1981). *Robust Statistics*. John Wiley & Sons, New York.
- Kusuoka, S. (2001). On law invariant coherent risk measures. *Advances in Mathematical Economics*, 3, 83–95.
- Merton, R. C. (1973). Theory of rational option pricing. *Bell Journal of Economics and Management Science*, 4(1), 141–183.
- Morgan Guaranty Trust Company (1995). *RiskMetrics Technical Document* (4th ed.). J.P. Morgan, New York.
- Rockafellar, R. T., and Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3), 21–41.
- Wang, T. (1996). *A Characterization of Dynamic Risk Measures*. Working paper, Faculty of Commerce, University of British Columbia.